

Gauge-Free Observable-Sector Extraction, Detector-Local Admissibility, and the Limits of Cross-Detector Geometry Identity in LIGO O4a H1–L1 Strain Data

Matěj Rada

Licensed under CC BY-NC-ND 4.0

July 30, 2026

Abstract

We construct and test a gauge-free primitive observable sector on two independent 4096 s LIGO O4a strain records from Hanford H1 and Livingston L1, each sampled at 16384 Hz and partitioned into 1024 non-overlapping four-second windows. The common observable map contains nine primitive coordinates: three log amplitude statistics and six log band powers. Duplicate and deliberately derived placement coordinates are excluded from the rank test. On H1, the relative spectrum of the empirical covariance proxy is 1, 1.027×10^{-1} , 6.688×10^{-2} , 5.289×10^{-2} , 4.926×10^{-2} , 4.105×10^{-2} , 2.861×10^{-2} , 2.784×10^{-2} , and 1.117×10^{-3} ; hence the primitive map has numerical rank nine at relative threshold 10^{-3} and rank eight at 10^{-2} . The same nine-coordinate construction has rank nine and zero thresholded hole dimension on L1 at 10^{-3} . This supports detector-local portability of the admissibility rule at the stated scale. It does not imply equality of the induced empirical geometries. Under a deliberately hostile direct index identification of the two non-contemporaneous records, the pairwise distance-geometry correlation is 0.1581447, and the nine sample-space principal angles range from 1.1546 to 1.5470 rad. Thus unconditional H1–L1 geometry identity is not supported. Finally, an H1 quadratic spectral placement functional is evaluated after sector extraction. It is an exactly reconstructible post-map functional and not a reconstruction of the Einstein tensor or physical stress-energy from a single detector. The combined result identifies the defensible object as a detector-portable admissibility class, together with an explicit still-missing response-aware transport required for any detector-independent physical geometry claim.

1 Scope and Logical Separation

The analysis separates four claims that must not be conflated:

- (i) *primitive sector construction*: a fixed gauge-free observable recipe is applied to each detector record;
- (ii) *detector-local admissibility*: the recipe passes a stated rank/null gate on each record;
- (iii) *cross-detector geometry identity*: the empirical geometries agree under a specified identification or transport;
- (iv) *physical placement*: a functional of the accepted observables is evaluated after extraction.

The data support (i), support (ii) at relative threshold 10^{-3} for the tested records, reject the simplest unconditional form of (iii), and support only a controlled proxy interpretation of (iv). In particular, equal rank does not establish equal covariance geometry, and a quadratic strain functional does not by itself yield a physical stress-energy tensor.

2 Data and Windowing

Two independent O4a strain records are used, one from H1 and one from L1. Each record has duration 4096 s and sampling frequency $f_s = 16384$ Hz, corresponding to 67,108,864 samples per detector. Each series is partitioned into 1024 non-overlapping intervals I_j of duration 4 s. The records are not asserted to be contemporaneous and are not treated as observations of the same physical strain realization.

Table 1: Input records and common observable construction.

Quantity	H1	L1
Sampling frequency	16384 Hz	16384 Hz
Duration	4096 s	4096 s
Samples	67,108,864	67,108,864
Window length	4 s	4 s
Non-overlapping windows	1024	1024
Primitive dimension	9	9

3 Gauge-Free Primitive Observable Map

Definition 3.1 (Primitive observable map). *For a four-second window I_j , let*

$$W_j = (\log \sigma_j, \log \|h\|_{\infty,j}, \log \kappa_j, \\ \log P_{20,80,j}, \log P_{80,300,j}, \log P_{300,1000,j}, \\ \log P_{1000,2048,j}, \log P_{2048,4096,j}, \log P_{4096,8192,j}),$$

where σ_j is the window standard deviation, $\|h\|_{\infty,j}$ is the maximum absolute strain, κ_j is kurtosis, and $P_{a,b,j}$ is the spectral power assigned to the frequency band $[a, b]$ Hz. The same estimator conventions are used for both detector records.

Logarithms require positive arguments. Therefore the computational domain consists of windows for which all nine primitive quantities are finite and strictly positive. Any clipping, flooring, or omitted-window rule used by the implementation is part of the estimator specification and must remain identical across detectors.

Definition 3.2 (Primitive-coordinate gate). *Duplicate coordinates and observables introduced solely as exact post-map summaries, including band-power ratios, spectral entropy computed from the chosen bands, and the quadratic placement functional below, are excluded from the primitive rank basis. They may be evaluated after the primitive sector has been accepted.*

Remark 3.1 (Algebraic dependence and covariance rank). *If an appended coordinate satisfies $z = g(U)$, then the augmented image $(U, g(U))$ lies on a lower-dimensional graph and its differential has no additional intrinsic tangent dimension. Exact singularity of a finite-sample covariance matrix follows automatically only for affine dependence, not for an arbitrary nonlinear g . Consequently, the exclusion of derived coordinates is a coordinate-design rule; a small covariance eigenvalue is evidence of numerical or approximately linear degeneracy at the sampled scale, not by itself a theorem of physical blindness.*

4 Empirical Rank and Thresholded Sectors

Let $X_D \in \mathbb{R}^{1024 \times 9}$ be the centered feature matrix for detector $D \in \{H1, L1\}$. Let C_D be the covariance matrix used by the implementation. If the features are additionally standardized to

unit marginal variance, C_D is the correlation matrix. The normalization choice must be fixed before comparing numerical spectra. Write

$$\lambda_{D,1} \geq \cdots \geq \lambda_{D,9} \geq 0$$

for its eigenvalues and $v_{D,i}$ for corresponding eigenvectors.

Definition 4.1 (Thresholded resolved and null sectors). *For relative threshold $\epsilon > 0$, define*

$$\mathcal{R}_{D,\epsilon} = \text{span}\{v_{D,i} : \lambda_{D,i}/\lambda_{D,1} > \epsilon\}, \quad \mathcal{N}_{D,\epsilon} = \mathcal{R}_{D,\epsilon}^\perp,$$

and

$$\text{rank}_\epsilon(D) = \dim \mathcal{R}_{D,\epsilon}.$$

The sector $\mathcal{N}_{D,\epsilon}$ is a thresholded empirical null sector, not an exact kernel unless the corresponding eigenvalues vanish.

Proposition 4.1 (H1 primitive spectrum). *For the tested H1 record, the relative eigenvalues are*

$$(1, 1.027 \times 10^{-1}, 6.688 \times 10^{-2}, 5.289 \times 10^{-2}, \\ 4.926 \times 10^{-2}, 4.105 \times 10^{-2}, 2.861 \times 10^{-2}, 2.784 \times 10^{-2}, 1.117 \times 10^{-3}).$$

Therefore $\text{rank}_{10^{-3}}(H1) = 9$ and $\text{rank}_{10^{-2}}(H1) = 8$.

Proof. All nine relative eigenvalues exceed 10^{-3} , while exactly eight exceed 10^{-2} under the strict threshold convention in the definition. $\square$

Proposition 4.2 (Detector-local portability at the stated threshold). *For the tested records,*

$$\text{rank}_{10^{-3}}(H1) = \text{rank}_{10^{-3}}(L1) = 9, \quad \dim \mathcal{N}_{H1,10^{-3}} = \dim \mathcal{N}_{L1,10^{-3}} = 0.$$

Thus the same primitive construction passes the stated rank/null gate on both detectors.

Proof. Direct eigendecomposition of each detector-local empirical covariance proxy gives nine relative eigenvalues strictly above 10^{-3} . $\square$

Table 2: Detector-local admissibility at relative threshold 10^{-3} .

Quantity	H1	L1
Primitive dimension	9	9
Rank at 10^{-3}	9	9
Thresholded hole dimension at 10^{-3}	0	0

Corollary 4.1 (Supported admissibility claim). *At resolution $\epsilon = 10^{-3}$, these records support detector-local extraction of the same nine-dimensional primitive observable sector. They do not support a nontrivial primitive null-sector claim at that threshold.*

Remark 4.1 (Threshold dependence). *The H1 rank changes from nine at 10^{-3} to eight at 10^{-2} . Full rank is therefore not a threshold-free statement. The smallest H1 direction is resolved only within the interval of relative thresholds below 1.117×10^{-3} , subject to sampling and estimator uncertainty. No biological or physical interpretation is assigned to that direction by the rank computation alone.*

5 H1 Effective-Rank and Local Closure Diagnostics

For the H1 spectrum, the participation rank and entropy rank were computed as

$$r_{\text{part}} = 1.835, \quad r_{\text{ent}} = 2.967.$$

These effective ranks quantify spectral concentration even though the thresholded algebraic rank is nine. They show that variance is strongly anisotropic: full thresholded rank does not mean that all directions carry comparable empirical weight.

The median principal angle between adjacent 256 s H1 resolved sectors was reported as 7.76×10^{-16} rad. At $\epsilon = 10^{-3}$, however, each accepted resolved sector is the full nine-dimensional ambient primitive feature space. The zero-angle result is consequently a closure check for full-span extraction, not evidence of nontrivial basis-vector or covariance stability. Nontrivial transport stability would require comparison of a proper leading subspace, spectral projectors associated with isolated eigenvalue clusters, or response-aware geometries.

Table 3: H1 diagnostics and their supported interpretation.

Quantity	Result
Rank at 10^{-2}	8
Rank at 10^{-3}	9
Participation rank	1.835
Entropy rank	2.967
Median adjacent 256 s full-span angle	7.76×10^{-16} rad
Placement log-identity error	6.73×10^{-17}

6 Cross-Detector Geometry Attack

The H1 and L1 records are neither assumed contemporaneous nor assumed to contain the same source realization. Direct identification of window j in H1 with window j in L1 is therefore not physically canonical. It can nevertheless be used as a hostile test of an especially strong claim: that raw empirical geometry is detector- and epoch-independent without calibration or learned transport.

Let S_D denote the resolved score matrix produced from X_D . Pairwise Euclidean distances between rows define a sample distance matrix D_D . Using the same direct window-index ordering for both records, the observed distance-geometry correlation is

$$\text{corr}(\text{vec}_{<}(D_{H1}), \text{vec}_{<}(D_{L1})) = 0.1581447,$$

where $\text{vec}_{<}$ extracts one off-diagonal triangle. Under the same index identification, the nine principal angles between the score-column subspaces in sample space range from

$$1.1546 \text{ rad} \quad \text{to} \quad 1.5470051 \text{ rad}.$$

Proposition 6.1 (Failure of direct unmatched-record identity). *For the tested unmatched records, direct window-index identification does not support equality of the H1 and L1 sample distance geometries or equality of their score-column subspaces.*

Proof. Under the stated identification, the distance-geometry correlation is 0.1581447, far from unity, and all reported sample-space principal angles are large, with the maximum close to $\pi/2$. These computed diagnostics are inconsistent with direct equality under that identification. $\square$

Remark 6.1 (What the attack does not test). *Because window indices do not label matched physical events, the attack does not estimate coherent H1–L1 signal agreement and does not rule out a common source manifold after time-delay correction, antenna-response modeling, calibration, whitening, event conditioning, or learned domain alignment. Its conclusion is narrower: full local rank does not supply an unconditional cross-detector isomorphism.*

7 Admissibility Class and Response-Aware Transport

Definition 7.1 (Detector-local admissibility predicate). *For a fixed feature recipe, estimator convention, preprocessing rule, and threshold ϵ , define*

$$\text{Adm}_D(W; \epsilon) = 1 \iff W \text{ passes the prescribed finiteness, primitive-coordinate, and rank/null gates on } D.$$

Two coordinate realizations belong to the same admissibility class only relative to explicitly stated allowed reparameterizations and gates.

Theorem 7.1 (Admissibility portability without geometry identity). *Let W be one fixed primitive construction applied separately to datasets D_1 and D_2 . Equality*

$$\text{Adm}_{D_1}(W; \epsilon) = \text{Adm}_{D_2}(W; \epsilon) = 1$$

does not imply $\mathcal{G}_{D_1} \cong \mathcal{G}_{D_2}$ for the induced empirical geometries. In the present computation, the antecedent holds at $\epsilon = 10^{-3}$ for H1 and L1, while direct unmatched-record geometry identity is not supported.

Proof. The admissibility predicate records a detector-local gate, here numerical rank and thresholded nullity together with primitive-coordinate validity. It contains no condition equating eigenvalues, eigenspaces, pairwise sample distances, or detector responses. Hence the implication to geometry identity is invalid in general. The H1–L1 computations instantiate this distinction: both local ranks are nine, whereas the direct cross-record distance correlation and sample-space angles disagree strongly with identity. $\square$

Corollary 7.1 (Necessary additional structure). *A defensible detector-independent physical geometry claim requires a specified response-aware map*

$$\mathcal{A}_{H1 \rightarrow L1} : \mathcal{G}_{H1} \longrightarrow \mathcal{G}_{L1}$$

and an acceptance criterion evaluated on data for which the comparison is physically meaningful. Candidate ingredients include contemporaneous common-source windows, inter-site time-delay correction, detector antenna responses, calibration transfer functions, common preprocessing, and out-of-sample validation. Such a map is not determined by rank equality.

8 Post-Map GR Placement Bridge

After primitive-sector extraction, define on each H1 window

$$\rho_{\text{obs}}(I_j) = \sum_b (2\pi f_b)^2 P_b(I_j),$$

where f_b is the declared representative frequency, here the geometric center, of band b . This functional uses only accepted band powers and is therefore a deterministic post-map coordinate.

Proposition 8.1 (Exact post-map reconstructibility). *Given the six primitive log band powers and the fixed band centers, ρ_{obs} is uniquely determined by*

$$\rho_{\text{obs}}(I_j) = \sum_b (2\pi f_b)^2 \exp(\log P_{b,j}).$$

The reported H1 log reconstruction discrepancy, 6.73×10^{-17} , is consistent with numerical roundoff in this algebraic identity.

Proof. Exponentiation recovers each positive band power from its primitive log coordinate, after which the displayed weighted sum is deterministic. No inverse physical field equation is solved. $\square$

Remark 8.1 (Physical status). *The notation ρ_{obs} denotes a quadratic spectral strain proxy. Overall normalization, detector response, gauge-invariant gravitational-wave energy definitions, sky position, polarization, and source reconstruction are not supplied by this identity. The bridge is therefore a placement functional within the observable representation, not a reconstruction of $G_{\mu\nu}$ or $T_{\mu\nu}$ from one interferometer.*

9 Figures

If the corresponding reproducibility artifacts are present, the H1 primitive spectrum and H1 adjacent-segment closure diagnostic may be included as follows.

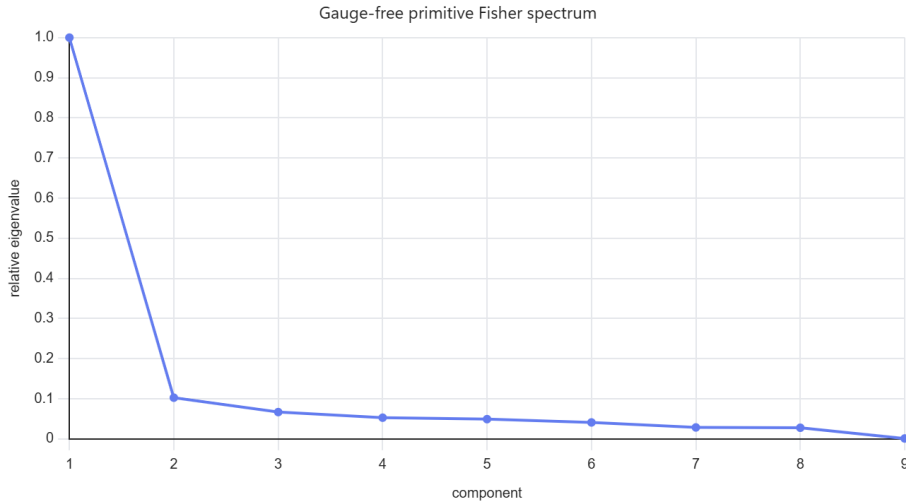


Figure 1: H1 gauge-free primitive covariance spectrum. The horizontal threshold, if plotted, must be interpreted relative to the largest eigenvalue.

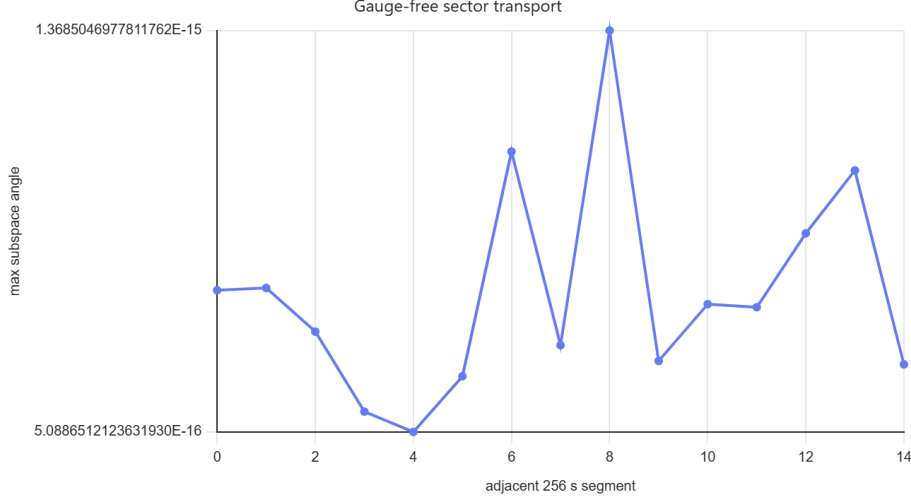


Figure 2: Adjacent 256 s H1 full-span angle diagnostic. Since the accepted sector is the full primitive space at 10^{-3} , this is a closure check rather than a nontrivial transport result.

10 Supported Claims and Non-Claims

Supported claims

- (1) The nine-coordinate primitive map is numerically full rank on the tested H1 and L1 records at relative threshold 10^{-3} .
- (2) The same construction therefore passes the stated detector-local rank/null admissibility gate on both records.
- (3) H1 is rank eight at 10^{-2} , demonstrating explicit resolution dependence.
- (4) Direct index-based comparison of the unmatched records does not support unconditional equality of their empirical distance geometry or sample-space score subspaces.
- (5) The quadratic spectral placement proxy is exactly computable from the primitive band-power coordinates.

Non-claims

This study does not claim gravitational-wave detection; a common event in the two records; equality of raw H1 and L1 score spaces; threshold-independent rank; a physical primitive blind sector at 10^{-3} ; uniqueness of the primitive coordinate formula; detector-independent covariance geometry; gauge-invariant gravitational-wave energy recovery; reconstruction of a source manifold, Einstein tensor, or physical stress-energy tensor; or validation of a response-aware H1–L1 transport. It also does not interpret large principal angles from unmatched windows as evidence against coherent astrophysical signals.

11 Reproducibility

The reported computational artifacts are

- `rg_o4a_h1_gaugefree_results.json`,
- `rg_o4a_h1_gaugefree_summary.txt`,

- `run_rg_o4a_h1_gaugefree.py`,
- `rg_o4a_h1_fullstory_features_4s.csv`, and
- `rg_h1_l1_transport_comparison.json`.

The inputs are the H1 and L1 4096 s O4a 16 kHz HDF5 strain files used by the computation. Exact reproduction additionally requires the input file names or stable identifiers, checksums, strain dataset paths, estimator definitions, preprocessing and finite-value rules, covariance normalization, eigensolver convention, and software environment to be recorded by the accompanying script or metadata. The strict inequality in the rank gate and the threshold $\epsilon = 10^{-3}$ are part of the specification.

12 Conclusion

The combined H1–L1 analysis supports an asymmetric but coherent conclusion. A common nine-dimensional gauge-free primitive observable recipe survives detector replacement at the stated numerical resolution: both records have rank nine and zero thresholded hole dimension at 10^{-3} . This establishes portability of a detector-local admissibility predicate, not universality of the resulting empirical geometry. When the unmatched records are forcibly identified by window index, their distance geometries correlate only weakly and their sample-space score subspaces are separated by large principal angles. The strong raw-geometry identity claim is therefore unsupported, exactly where detector, calibration, noise, and epoch dependence should be expected. The H1 quadratic spectral functional remains valid as a deterministic post-map placement proxy, but carries no stress-energy reconstruction claim. The next mathematical and computational object is consequently explicit: a response-aware, physically paired, out-of-sample validated transport between detector-conditioned admissible geometries.